

Thus, as a first estimate, the **longitudinal derivatives with respect to translational acceleration** can be approximated as

$$\begin{aligned}
 F_{x_b, \dot{u}} &\cong F_{z_b, \dot{u}} \cong M_{y_b, \dot{u}} \cong F_{x_b, \dot{w}} \cong 0 \\
 F_{z_b, \dot{w}} &\equiv \frac{\partial F_{z_b}}{\partial \dot{w}} = \frac{1}{V_o} \frac{\partial F_{z_b}}{\partial \dot{\alpha}} = -\frac{\frac{1}{2} \rho V_o^2 S_w}{V_o} \frac{\partial C_L}{\partial \dot{\alpha}} \\
 &\cong -\eta_h \frac{\rho S_w S_h l_{wt}}{\pi b_w^2} \frac{\partial C_{L_w}}{\partial \alpha} \frac{\partial C_{L_h}}{\partial \alpha} \\
 M_{y_b, \dot{w}} &\equiv \frac{\partial M_{y_b}}{\partial \dot{w}} = \frac{1}{V_o} \frac{\partial M_{y_b}}{\partial \dot{\alpha}} = \frac{\frac{1}{2} \rho V_o^2 S_w \bar{c}_w}{V_o} \frac{\partial C_m}{\partial \dot{\alpha}} \\
 &\cong \eta_h \frac{\rho S_w S_h l_{wt} x_{bh}}{\pi b_w^2} \frac{\partial C_{L_w}}{\partial \alpha} \frac{\partial C_{L_h}}{\partial \alpha} \\
 l_{wt} &= \begin{cases} 1.1(x_{b \text{ wingtip}} - x_{bh}), & x_{b \text{ wingtip}} > x_{bh} \\ 0.0, & x_{bh} > x_{b \text{ wingtip}} \end{cases}
 \end{aligned} \tag{7.6.66}$$

As mentioned previously, all of the **lateral derivatives with respect to translational acceleration are either zero or negligible**.

All of the aerodynamic derivatives that appear in the linearized equations of motion are complex functions of the shape and placement of the wing, tail, fuselage, and engines. For any given aircraft, these derivatives also vary with airspeed and other operating conditions. In general, these derivatives will need to be determined either from computer simulations or from wind tunnel tests. However, as a first estimate, the approximations given in this section may be used.

EXAMPLE 7.6.1. A large jet transport is flying at 30,000 feet with a steady level airspeed of 450 mph, which is the minimum drag airspeed for this altitude. This airplane has the following properties in cruise configuration:

$$\begin{aligned}
 S_w &= 5,500 \text{ ft}^2, \quad b_w = 196 \text{ ft}, \quad \bar{c}_w = 28 \text{ ft}, \quad x_{bw} = 0, \quad C_{L_w, \alpha} = 4.67, \\
 S_h &= 1,300 \text{ ft}^2, \quad x_{bh} = -100 \text{ ft}, \quad x_{b \text{ wingtip}} = -50 \text{ ft}, \quad l_{wt} = 55 \text{ ft}, \quad C_{L_h, \alpha} = 3.50, \\
 W &= 636,600 \text{ lbf}, \quad C_{L, \alpha} = 5.50, \quad C_{m, \alpha} = -1.26
 \end{aligned}$$

The maximum lift-to-drag ratio is 14. Assuming constant thrust aligned with the direction of flight and the center of gravity, estimate all dimensional longitudinal aerodynamic derivatives that appear in the homogeneous form of Eq. (7.5.23).

Solution. For this airplane the airspeed and the Mach number are

$$V_o = 450 \frac{5,280}{3,600} \text{ ft/sec} = 660 \text{ ft/sec}$$